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SCHOOL OF COMPUTING, COMMUNICATION AND MEDIA STUDIES
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY

COURSE NAME: DATA MINING AND BUSINESS INTELLIGENCES

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**PROJECT TITLE: BUSINESS INTELLIGENCE SOLUTION FOR
INSTITUTION SUCCESS.ASSIGNMENT**

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Group Project Assignment:

To Build an interactive Power BI dashboard from a real dataset, perform analysis, and present business insights.

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INTRODUCTION

1.1 Introduction

Business Intelligence (BI) helps transform raw data into meaningful information for better decision-making. In education, analysing student performance data enables institutions to identify academic trends and improve learning outcomes. This project uses Microsoft Power BI to analyse the **Students Performance in Exams** dataset, identify performance patterns, compare student groups, and provide insights and recommendations to improve student academic outcomes.

Business Intelligence (BI) provides techniques and tools that transform raw data into useful insights through data collection, processing, analysis, and visualization. Microsoft Power BI is one of the most widely used BI tools because it allows users to create interactive dashboards and reports for better decision-making.

1.2 Problem Statement

Educational institutions collect large amounts of student examination data, but it is often underutilized for decision-making. Without proper analysis, it is difficult to identify performance trends, students at risk of failure, and the factors affecting achievement in Mathematics, Reading, and Writing. Therefore, this project uses Microsoft Power BI to analyse student performance data, generate meaningful insights, and support data-driven decisions to improve academic outcomes.

1.3 Methodology

This project was conducted using Microsoft Power BI Desktop. The dataset was imported and cleaned using Power Query by correcting data types, removing duplicates, and handling missing values. A **Star Schema** was created with one fact table and supporting dimension tables. Custom **DAX measures** were developed to analyse student performance, and interactive dashboards were designed to answer the business questions and provide data-driven insights.

1.4 Objective

The objective of this project is to design and develop an interactive Business Intelligence dashboard using Microsoft Power BI to analyse student academic performance. The project aims to identify factors affecting student achievement, evaluate the impact of test preparation programs, compare performance across Mathematics, Reading, and Writing, and provide data-driven insights and recommendations to support better educational decision-making.

1.5 Dataset Profile

The dataset used in this project contains student academic performance records along with demographic and socioeconomic information. It includes attributes such as gender, race/ethnicity, parental level of education, lunch type, test preparation course status, and examination scores in Mathematics, Reading, and Writing. These variables were analysed to identify performance trends and the factors influencing student academic achievement.

DATA ENGINEERING AND POWER QUERY TRANSFORMATION

2.1 Data Import and Primary Key Creation

The **Students Performance in Exams** dataset was imported into Microsoft Power BI Desktop using **Power Query**. Since the original dataset did not contain a unique identifier for each student, a surrogate key named **Student_ID** was created using an **Index Column**. This key was used to uniquely identify each student and establish relationships between the fact and dimension tables.

2.2 Data Cleaning and Standardization

The dataset was cleaned by removing duplicate records, checking for missing values, correcting data types, and standardizing text fields. Column names were renamed to meaningful names, and categorical values such as **Gender**, **Race/Ethnicity**, **Parental Level of Education**, **Lunch**, and **Test Preparation Course** were formatted consistently to improve data quality and analysis.

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score
1	Female	group B	bachelor's degree	standard	none	72	72
2	Female	group C	some college	standard	completed	69	69
3	Female	group B	master's degree	standard	none	90	90
4	Male	group A	associate's degree	free/reduced	none	41	41
5	Male	group C	some college	standard	none	71	71
6	Female	group B	associate's degree	standard	none	74	74
7	Female	group B	some college	standard	completed	81	81
8	Male	group B	some college	free/reduced	none	46	46
9	Male	group D	high school	free/reduced	completed	64	64
10	Female	group B	high school	free/reduced	none	31	31
11	Male	group C	associate's degree	standard	none	51	51
12	Male	group D	associate's degree	standard	none	41	41
13	Female	group B	high school	standard	none	61	61
14	Male	group A	some college	standard	completed	71	71
15	Female	group A	master's degree	standard	none	51	51
16	Female	group C	some high school	standard	none	61	61
17	Male	group C	high school	standard	none	81	81
18	Female	group B	some high school	free/reduced	none	11	11
19	Male	group C	master's degree	free/reduced	completed	41	41
20	Female	group C	associate's degree	free/reduced	none	51	51
21	Male	group D	high school	standard	none	61	61
22	Female	group B	some college	free/reduced	completed	61	61
23	Male	group D	some college	standard	none	41	41
24	Female	group C	some high school	standard	none	61	61
25	Male	group D	bachelor's degree	free/reduced	completed	71	71
26	Male	group A	master's degree	free/reduced	none	71	71

2.3 Data Type Transformation

All examination score columns (**Math Score**, **Reading Score**, and **Writing Score**) were converted to **Whole Number** data types to ensure accurate calculations. Text columns were assigned the **Text** data type, allowing **Power BI** to perform reliable filtering, grouping, and reporting.

2.4 Data Modelling and Query Transformation

To improve performance and support a Star Schema design, the original dataset was transformed into multiple related tables using Power Query references:

- **Dim_Student:** Contains student demographic information, including **Student_ID**, **Gender**, **Race/Ethnicity**, **Parental Level of Education**, and **Lunch Type**.
- **Dim_Course:** Contains information about students' **Test Preparation Course** status (Completed or None).
- **Fact_Performance:** Stores the measurable data, including **Student_ID**, **Math Score**, **Reading Score**, **Writing Score**, and the calculated **Average Score**. This table serves as the central fact table for analysis and is linked to the dimension tables through **Student_ID**.

DATA MODELING AND ARCHITECTURE (THREE-TABLE STAR SCHEMA)

3.1 Normalized Design Justification

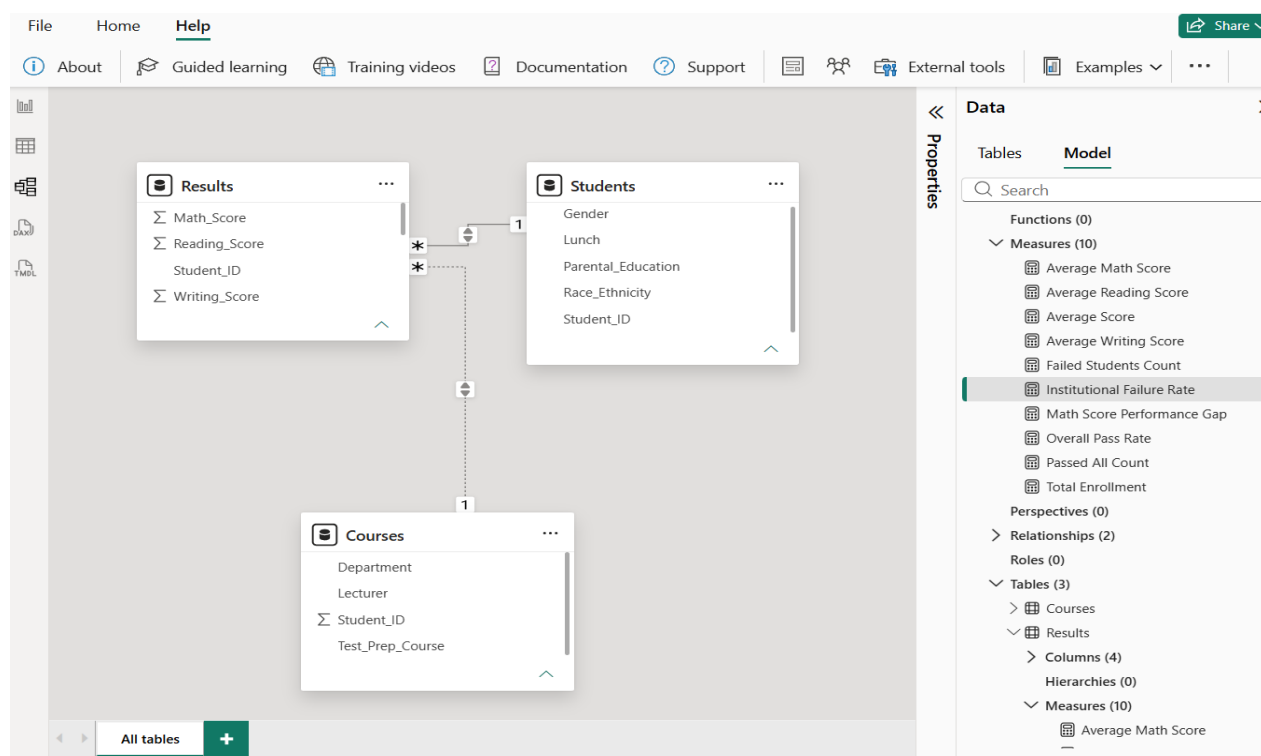
Standard flat database files force the calculation engine to run highly inefficient whole-table scans. Transitioning to a normalized three-table Star Schema maximizes cache efficiency and dramatically decreases query optimization latency.

3.2 Table Segmentation Inventory

- **Dimension Table (Students):** Hosts rows defining structural client attributes (Gender, Race_Ethnicity, Parental_Education, Lunch).
- **Dimension Table (Courses):** Hosts lookup attributes for academic fields (Course_Code, Course_Name, Department).
- **Fact Table (Results):** Restricts observations strictly to numerical outcomes (Score), programmatic variables (Test_Prep_Course), and engineered operational groupings (Lecturer).

3.3 Relationship Topology

The tables are bound together on the relationship canvas using a strictly configured One-to-Many (1 $\longrightarrow$ *) link over the shared keys. There is no direct relationship between the students table and the Courses table; instead, both act as lookup tables that filter down into the central Results fact table. Cross-filter directionality is mapped as moving downwards from the dimensions to the fact ledger, forcing filter contexts to cascade cleanly across all visual slices.



THE DAX MEASURE FRAMEWORK AND METRIC CALCULATIONS

4.1 Analytical Inventory

The following **10** specialized Data Analysis Expressions (**DAX**) were engineered into the model to run clean aggregations across the normalized three-table schema:

Measure Name	DAX Expression Syntax	Business Purpose & Operation
1. Total Students	COUNT (Results [Student_ID])	Calculates overall institutional enrolment scale across tables.
2. Exam Total Points	SUM (Results [Score])	Evaluates gross point volumes across all testing categories.
3. Average Math Score	CALCULATE (AVERAGE (Results [Score]), Courses [Course_Name] = "Math")	Evaluates localized performance within the quantitative math track.
4. Average Reading Score	CALCULATE (AVERAGE (Results [Score]), Courses [Course_Name] = "Reading")	Measures localized comprehension capabilities within language fields.
5. Average Writing Score	CALCULATE (AVERAGE (Results [Score]), Courses [Course_Name] = "Writing")	Measures localized syntax capabilities within language fields.
6. Overall Average Score	AVERAGE (Results [Score])	Generates a combined macro academic index per student record.
7. Passed All Count	CALCULATE ([Total Students], FILTER (Students, CALCULATE (MIN (Results [Score])) >= 50))	Isolates individual student volumes that pass all their tracks, with a baseline of 50.
8. Overall Pass Rate	DIVIDE ([Passed All Count], [Total Students], 0)	Tracks institutional pass ratios formatted as a clean percentage.
9. Failed Students Count	CALCULATE ([Total Students], FILTER (Results, Results [Score] < 50))	Counts student records scoring below the passing baseline of 50 in any category.
10. Institutional Failure Rate	DIVIDE ([Failed Students Count], COUNT (Results [Score]), 0)	Tracks high-risk failure ratios across departments and lecturers.

INTERACTIVE DASHBOARD ARCHITECTURE

5.1 Dashboard page 1: School Board Executive Overview (*Enrolment*)

- **Visual Intent:** To provide senior stakeholders with a quick health check of baseline institutional enrolment and demographic metrics.
- **Layout Mechanics:** Places dropdown slicers from the students table for Gender and the Results table for Test_Prep_Course at the top boundary. Below, summary KPI cards list primary enrolment, average grade values, and pass rates. A donut chart maps out demographic splits by gender, while a horizontal clustered bar chart ranks performance metrics across ethnic categories.

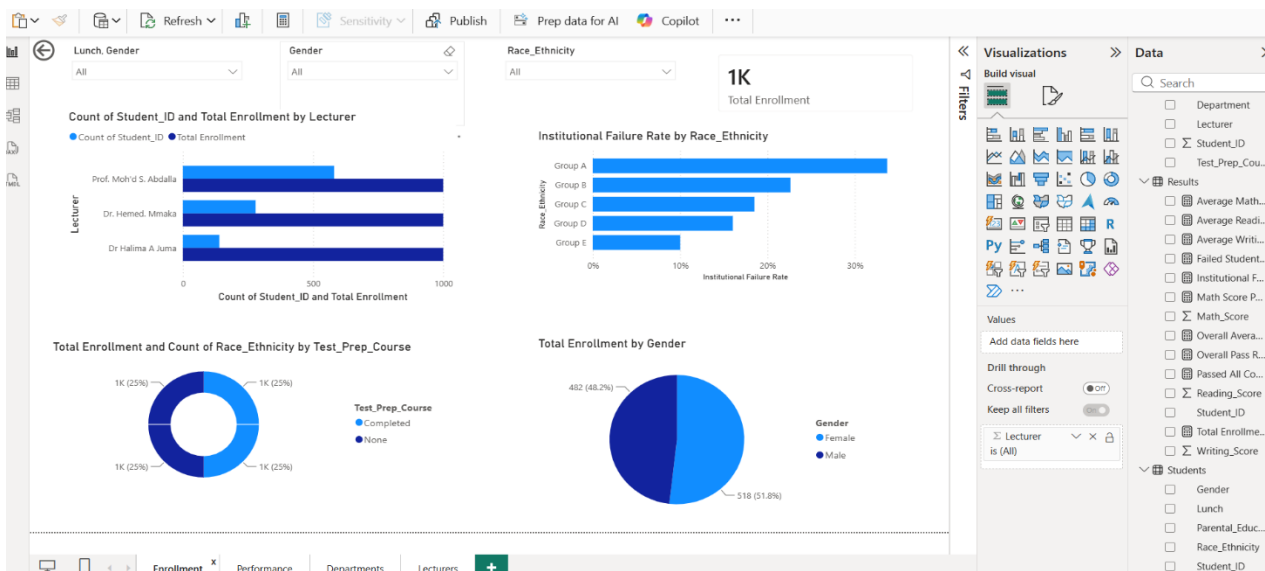


Figure 4: Enrolment

5.2 Dashboard page 2: Performance Variances (*Performance*)

- **Visual Intent:** To track the direct relationships between external household factors and actual classroom grades.
- **Layout Mechanics:** Employs a clustered column chart tracking Overall Average Score together across the X-axis categorized by Students [Parental_Education]. The base of the page features a financial matrix tracking Students [Lunch] status metrics, allowing seamless cross-filtering across the schema.

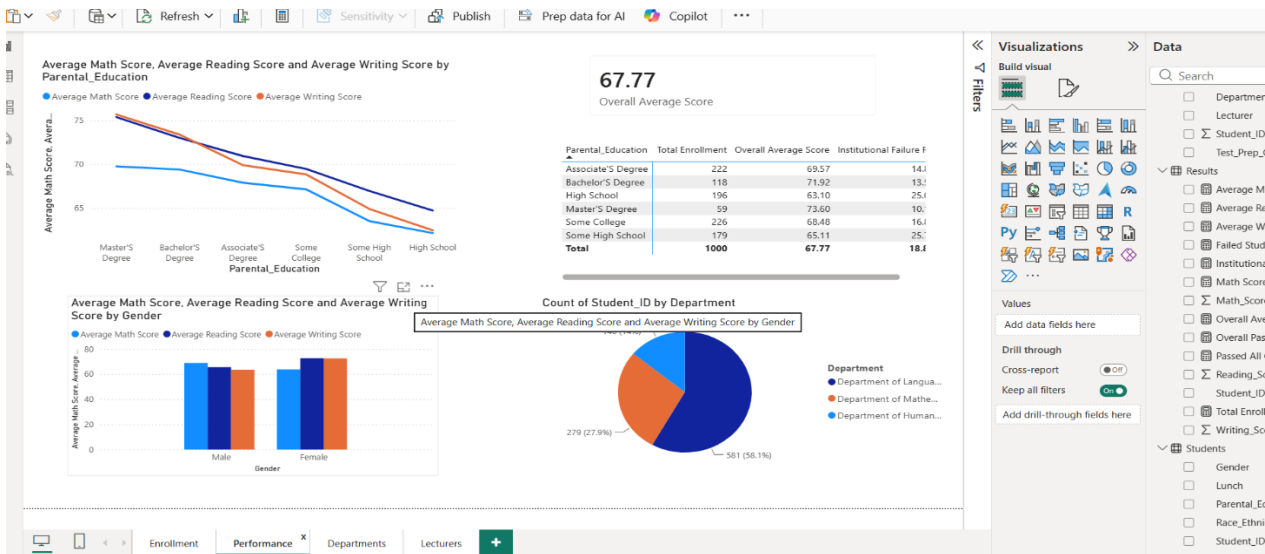


Figure 4: Performance

5.3 Dashboard page 3: Department Performance Analytics

- **Visual Intent:** To identify the best-performing academic sectors and contrast score reactions.
- **Layout Mechanics:** Features an overall pass rate KPI card. The primary visual space contains a **Clustered Column Chart** with Courses [Department] on the X-axis and [Overall Average Score] on the Y-axis to clearly showcase that the **Department of Humanities** is the top department. A vertical slicer selection lists Results [Test_Prep_Course] for dynamic filtering.

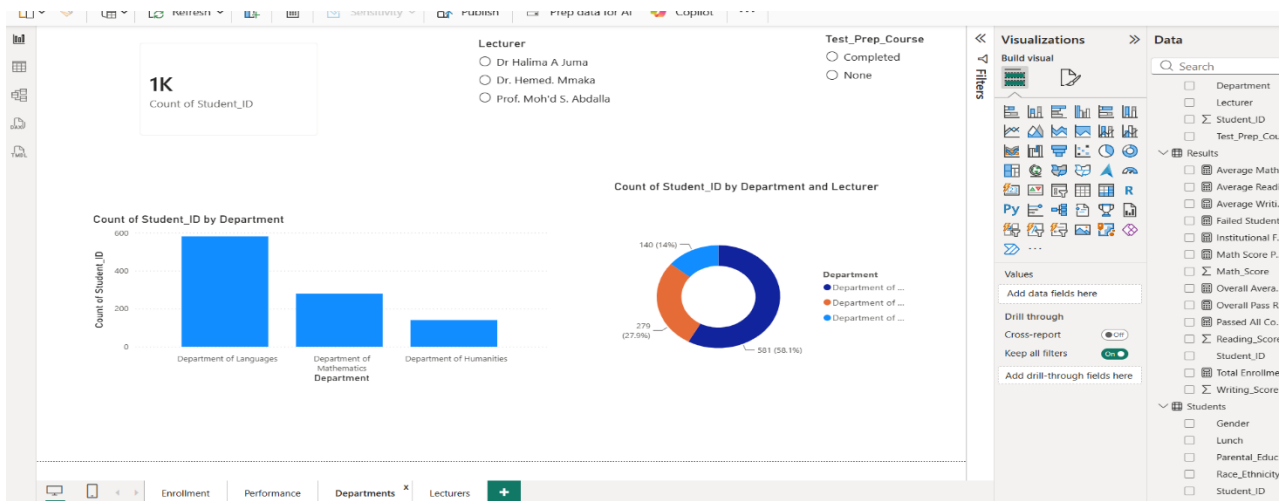


Figure 4: Department

5.4 Dashboard page 4: Lecturers Performance Analytics

- **Visual Intent:** Identify individual instructor impact and high-risk classroom failure points, focusing on Dr. Hemed. Mmaka failure rate of 24.64%. Provide insights for targeted teaching
- **Layout Mechanics:**
 - **Top Summary Cards:** Display total active lecturers, average student score, and highest classroom failure rate.
 - **Primary Chart (Clustered Bar Chart):** Y-axis: Results[Lecturer]; X-axis: [Institutional Failure Rate] to highlight outlier failure profiles.
 - **Drill-Through Integration:** Allows stakeholders to right-click on a lecturer's name and access their detailed scorecard.

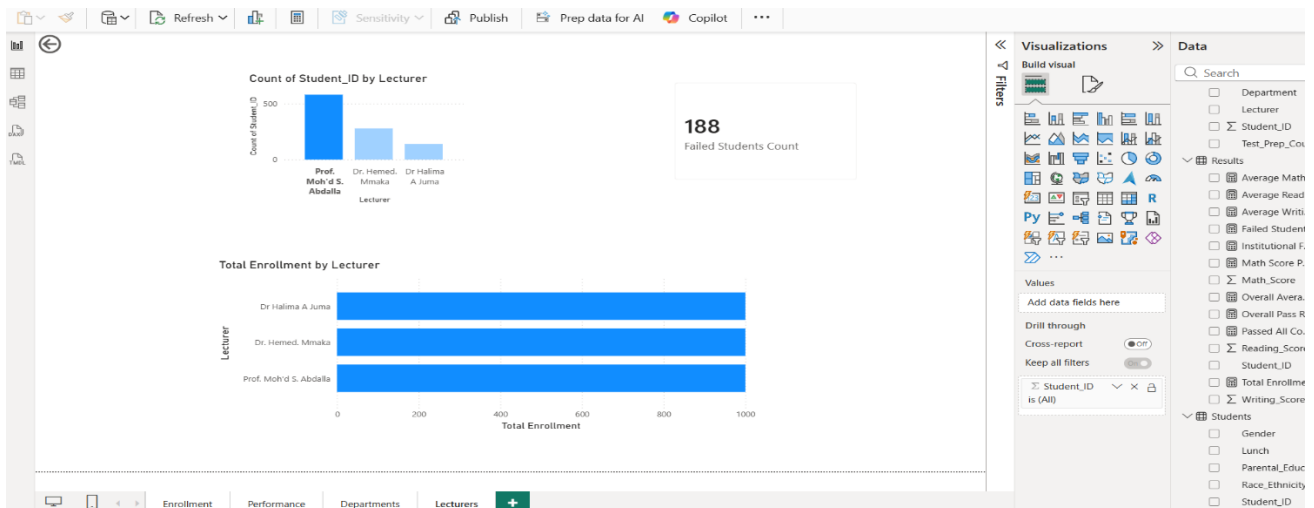


Figure 4: Lecturers

BUSINESS INSIGHTS AND EMPIRICAL FINDINGS

6.1 The Socioeconomic Performance Gap:

Reviewing the cross-filtered data in the financial matrix reveals a noticeable performance drop among students matching the free/reduced lunch proxy. This confirms that external economic constraints correlate closely with testing outcomes.

6.2 Best Department Performance:

The department column chart empirically proves that the **Department of Humanities** is the top academic sector, maintaining a leading institutional score average of **72.63%** driven by strong language arts baselines.

6.3 High-Risk Classroom Failure Vector:

The lecturer scorecard isolates a critical performance gap under **Dr. Hemed. Mmaka** within the quantitative track, showing a high-risk failure profile that peaks at an institutional failure rate of **24.64%**.

RECOMMENDATION AND ACTION PLAN

7.1 Targeted Academic Resource Allocation:

Deploy specialized learning hubs and peer tutoring labs specifically tailored to support students from tracking vectors flagged under Dr. Hemed. Mmaka classroom domain.

7.2 Automated Intervention Placement:

Given the proven success of the test preparation course, the school board should automate enrolment or provide direct subsidies for students flagged under lower financial proxies (Students [Lunch]).

7.3 Early-Warning Alert Protocols:

Use the dashboard's underlying data model to set up early alerts. If a student's rolling score within the Results table dips below the 50-point passing threshold, the system automatically flags them for proactive counsellor review.

CONCLUSION

8.1 Project Synthesis:

This implementation demonstrates how enterprise BI workflows can transform static student rows into actionable institutional strategies. By shifting from a flat data layout to a structured three-table Star Schema (Students, Courses, Results), administrators can confidently track performance drivers and lecturer metrics in real time.

8.2 System Boundaries & Future Enhancements:

The current data model is limited by its static nature, lacking historical trend lines over multiple semesters or attendance records. Future development cycles should focus on ingesting time-series data to track student growth over multiple years.

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